LOSARTAN

No cough (unlike ACEi)

Teratogenic

CHEMICAL NAME [2-butyl-5-chloro-3-({4-[2-(1H-tetrazol-5-yl)phenyl]phenyl}methyl)-3H-imidazol-4-yl]methanol (potassium)

CLASS Angiotensin II receptor blocker (ARB, sartan)

TRADE NAMES Cozaar · Lorsan · Hyzaar (with HCTZ) · Losar

THERAPEUTIC USES / INDICATIONS

- Hypertension (incl. with LV hypertrophy; stroke reduction).
- Diabetic nephropathy with proteinuria (type 2 diabetes).
- Heart failure when ACE inhibitor not tolerated (cough).
- Lowers uric acid modestly (useful in gout).

ADMINISTRATION

- **PO** once daily with or without food.

DOSAGES

- **Hypertension:** 50 mg once daily; may increase to 100 mg.
- Start 25 mg if volume-depleted or hepatic impairment.
- **Heart failure:** 12.5 mg daily, titrate to 150 mg.
- **Children ≥6 yr:** 20–50 kg 25 mg once daily (max 50 mg); >50 kg 50 mg (max 100 mg).

PHARMACOKINETICS

- **Onset:** 6 h (full effect 3–6 weeks). **Duration:** 24 h.
- First-pass metabolism (CYP2C9, 3A4) to active E-3174 metabolite (more potent).
- Excreted in bile and urine. **t½:** 2 h (metabolite 6–9 h).

PHARMACODYNAMICS

- Blocks **angiotensin II type 1 (AT1) receptor** → vasodilation, ↓ aldosterone.
- Does not inhibit bradykinin breakdown → no cough, less angioedema.
- Uricosuric effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, fatigue.
- Hypotension.
- Upper respiratory symptoms, back pain.

ADVERSE EFFECTS (SERIOUS)

- Hyperkalaemia.
- Acute kidney injury (renal artery stenosis, dehydration, NSAIDs).
- Angioedema (rare).
- Hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (2nd/3rd trimester).
- Use with aliskiren in diabetes or eGFR <60.
- **Caution:** bilateral renal artery stenosis, hepatic impairment, volume depletion.

KEY INTERACTIONS

- **K^+-sparing diuretics, supplements, trimethoprim:** hyperkalaemia.
- **NSAIDs:** ↓ effect, AKI.
- **Lithium:** toxicity.
- **ACE inhibitors:** dual RAAS blockade — avoid.

NURSING MANAGEMENT

- Monitor BP, U&E (K^+, creatinine).
- Check for dehydration before giving.
- Postural BP in elderly.
- Monitor proteinuria in diabetic nephropathy.
- Hold if SBP low or K^+ high.

PATIENT EDUCATION

- Take once daily; full effect in a few weeks.
- Rise slowly.
- Avoid NSAIDs and potassium salt substitutes.
- Stop and see doctor if pregnant.

KEY POINTS

- ARB; blocks AT1 receptor.
- Renoprotective in diabetes.
- No bradykinin cough — ACEi alternative.
- Hyperkalaemia; avoid in pregnancy.

SOURCES CHECKED **[276]** Losartan potassium 100 mg film-coated tablets (UK SmPC) **[277]** Losartan Potassium and Hydrochlorothiazide, oral (US label) **[278]** Losartan Potassium and Hydrochlorothiazide, oral (US label, Proficient Rx LP)

Full references p. 432–457